

Development of a Semi-Autonomous Navigation System for a Pipeline Monitoring Robot

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Abstract—Pipelines are critical for the long-distance transportation of oil and gas; however, their structural integrity is continuously threatened by corrosion, mechanical defects, and third-party interference. Traditional manual monitoring approaches are labor-intensive and hazardous. This paper presents the design and implementation of a semi-autonomous, Raspberry-Pi-based robotic system for external pipeline monitoring. The system integrates ultrasonic obstacle detection, an autonomous halt capability, and a web-based teleoperation interface utilizing a RESTful API architecture. Empirical performance evaluations demonstrate a 95.00% obstacle-detection accuracy, a mean autonomous halt response time of 1.02 seconds, and an IoT email-alert latency of 5.15 seconds. These results validate a highly responsive, cost-effective mobile IoT solution for safe remote pipeline inspection.

Index Terms—Pipeline monitoring, Internet of Things (IoT), Robotics, Teleoperation, Autonomous navigation

I. INTRODUCTION

Pipelines remain the cornerstone of global energy transportation due to their safety and cost-effectiveness. Despite these advantages, extensive pipeline networks are highly susceptible to environmental degradation, corrosion, and third-party vandalism [1]. Undetected mechanical defects can escalate into catastrophic leakages or explosions, resulting in severe ecological damage and economic loss. Traditional monitoring approaches, primarily manual patrolling, are slow, resource-intensive, and inherently hazardous for human operators [2].

While advances in robotics have introduced mobile inspection alternatives, the field is heavily dominated by complex internal pipeline robots (PIGs) or highly expensive SLAM-enabled (Simultaneous Localization and Mapping) crawlers. There remains a critical gap for cost-effective, external monitoring robots that bridge the divide between purely manual remote control and fully autonomous navigation.

This work addresses this gap by developing a semi-autonomous external pipeline-monitoring robot. By integrating a lightweight RESTful IoT communication framework with localized ultrasonic hazard detection, the system provides safe, remote human oversight while guaranteeing autonomous collision prevention.

II. RELATED WORK

The shift from manual inspection to robotic systems has transformed pipeline maintenance. Internal robotic systems

utilize sophisticated sensors like magnetic flux leakage (MFL) and ultrasonic transducers to detect internal defects, but frequently struggle with tight bends and varying pipe diameters [2]. External robots, conversely, traverse the outer surface to inspect for coating damage and physical force [3]. However, many external robotic solutions rely on complex locomotion mechanisms and lack integrated cloud connectivity.

The integration of the Internet of Things (IoT) enhances scalability by connecting distributed sensor nodes and microcontrollers to cloud platforms [4]. Recent studies have demonstrated the efficacy of IoT-enabled intelligent robots that transmit telemetry to centralized servers [5]. By pairing robust RESTful communication interfaces [6] with simple, low-latency obstacle detection algorithms [7], robotic platforms can achieve reliable semi-autonomous behavior without the computational overhead of advanced LiDAR-based mapping.

III. SYSTEM ARCHITECTURE

The robotic system consists of two primary subsystems: a physical hardware platform and a cloud-based teleoperation interface.

A. Hardware and Locomotion

The mobile platform is governed by a Raspberry Pi 4 Model B microcontroller. Locomotion is achieved via a custom chassis driven by dual 550 high-torque DC geared motors (83 RPM, 53 kg-cm torque). These actuators are controlled by an L298N H-Bridge driver, allowing the Raspberry Pi to execute precise directional commands and pulse-width modulation (PWM) speed adjustments. Power is supplied by a 14.8V lithium-ion battery pack stepped down via a DC-DC buck converter, while the logic board is independently powered to prevent voltage drops during motor stalls.

For environmental perception, an HC-SR04 ultrasonic sensor is mounted on the forward chassis to provide continuous, non-contact distance measurements.

B. IoT Communication Interface

The control interface relies on a React.js frontend and a Node.js/Express.js backend utilizing a MongoDB database. Communication between the operator and the robot is handled asynchronously via a RESTful API. JSON-formatted commands are transmitted via HTTP POST requests, which the Raspberry Pi retrieves through continuous polling.

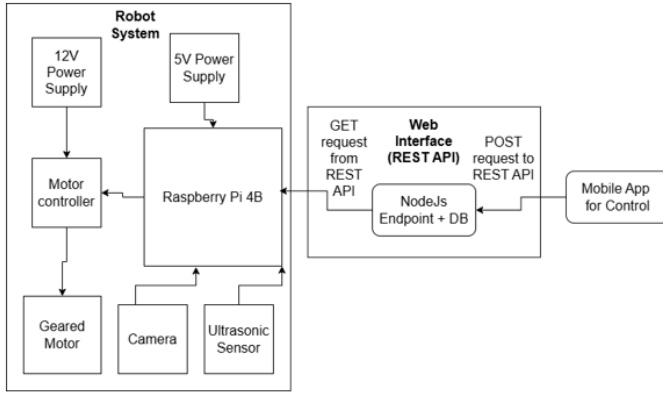


Fig. 1. System architecture integrating the robotic hardware and IoT API.

IV. METHODOLOGY

The system operates using a hybrid navigation logic designed to prioritize hardware safety over manual operator commands.

A. Manual Teleoperation

Under normal conditions, the robot polls the REST API endpoint to retrieve operator navigation commands (Forward, Backward, Stop). Upon executing a retrieved command, the robot posts a status update back to the API, closing the feedback loop and preventing command duplication.

B. Autonomous Hazard Prevention

To prevent collisions caused by network latency or operator error, the robot runs a concurrent autonomous safety loop. The ultrasonic sensor continuously calculates the time-of-flight of acoustic pulses to detect obstacles. If an object is detected within a critical 20 cm threshold, the localized logic overrides the API commands. The robot halts motor actuation immediately and triggers an automated SMTP email alert to the operator, holding its position until a reverse command is explicitly issued.

V. RESULTS AND DISCUSSION

The hardware was fully assembled on a wheeled chassis, and the REST API control loop was tested over standard cellular networks.

A. Performance Evaluation

The system's semi-autonomous reliability was quantified through a series of empirical trials measuring detection accuracy and network latency.

TABLE I
SYSTEM RESPONSE TIME AND ACCURACY METRICS

Performance Metric	Measured Value
Obstacle Detection Accuracy	95.00% (19/20 trials)
Mean Obstacle Detection Time	0.85 seconds
Mean Autonomous Halt Time	1.02 seconds
Mean Email Alert Latency	5.15 seconds

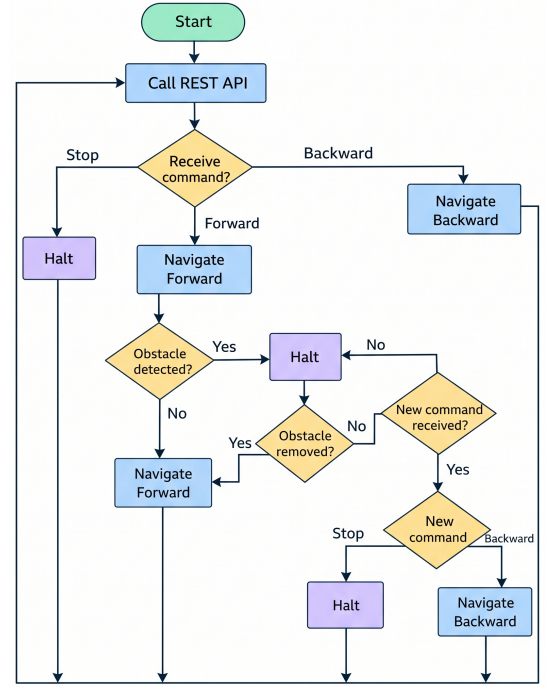


Fig. 2. Logic flowchart of the semi-autonomous navigation algorithm.



Fig. 3. The fully assembled semi-autonomous pipeline monitoring robot.

As detailed in Table I, the ultrasonic array successfully detected physical obstructions in 95% of trials. The critical safety metric—the autonomous halt time—averaged 1.02 seconds. Because this halt logic is executed locally on the Raspberry Pi's processor, it remains unaffected by remote network latency, ensuring immediate collision avoidance. The secondary IoT alert latency averaged 5.15 seconds, proving the RESTful architecture is highly capable of delivering near real-time hazard notifications to remote operators.

VI. CONCLUSION

A semi-autonomous navigation system for external pipeline monitoring was successfully developed and validated. By combining a robust Raspberry Pi hardware platform with a lightweight REST API and localized ultrasonic collision prevention, the system achieves a 95% hazard detection accuracy and rapid 1.02-second halt responses. This hybrid architecture presents a highly scalable, cost-effective alternative to purely manual inspection or complex SLAM-based robotics, ensuring operator safety and pipeline integrity.

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